

CHAPTER II.5

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5.1. Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -module of finite rank. We define the *dual* of $\mathcal{E}$ to be the sheaf $\mathcal{E}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$.

- (a) Show that $(\mathcal{E}^\vee)^\vee \cong \mathcal{E}$.
- (b) For any $\mathcal{O}_X$ -module $\mathcal{F}$, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}) \cong \mathcal{E}^\vee \otimes_{\mathcal{O}_X} \mathcal{F}$.
- (c) For any $\mathcal{O}_X$ -modules $\mathcal{F}, \mathcal{G}$,

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E} \otimes \mathcal{F}, \mathcal{G}) \cong \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{G})).$$

- (d) (*Projection Formula*). If $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, $\mathcal{F}$ is an $\mathcal{O}_X$ -module, and $\mathcal{E}$ is a locally free $\mathcal{O}_Y$ -module of finite rank, then there is a natural isomorphism

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{E}) \cong f_*(\mathcal{F}) \otimes_{\mathcal{O}_Y} \mathcal{E}.$$

Proof. □

5.3. Let $X = \text{Spec } A$ be an affine scheme. Show that the functors $\widetilde{}$ and Γ are adjoint. That is, for any A -module M , and for any sheaf of $\mathcal{O}_X$ -modules $\mathcal{F}$, there is a natural isomorphism

$$\text{Hom}_A(M, \Gamma(X, \mathcal{F})) \cong \text{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathcal{F})$$

Proof. □

5.5. Let $f : X \rightarrow Y$ be a morphism of schemes.

- (a) Show by example that if $\mathcal{F}$ is coherent on X , then $f_*\mathcal{F}$ need not be coherent on Y , even if X and Y are varieties over a field k .
- (b) Show that a closed immersion is a finite morphism.
- (c) If f is a finite morphism of noetherian schemes, and if $\mathcal{F}$ is coherent on X , then $f_*\mathcal{F}$ is coherent on Y .

5.6. *Support*. Recall the notions of support of a section of sheaf, support of a sheaf, and subsheaf with supports from (Ex. 1.14) and (Ex. 1.20).

- (a) Let A be a ring. Let M be an A -module, let $X = \text{Spec } A$, and let $\mathcal{F} = \widetilde{M}$. For any $m \in M = \Gamma(X, \mathcal{F})$, show that $\text{Supp } m = V(\text{Ann } m)$, where $\text{Ann } m = \{a \in A \mid am = 0\}$ is the *annihilator* of m .
- (b) Now suppose that A is noetherian, and M is finitely generated. Show that $\text{Supp } \mathcal{F} = V(\text{Ann } M)$.
- (c) The support of a coherent sheaf on a noetherian scheme is closed.
- (d) For any ideal $\mathfrak{a} \subset A$, we define a submodule $\Gamma_{\mathfrak{a}}(M)$ of M by

$$\Gamma_{\mathfrak{a}}(M) = \{m \in M \mid \exists n > 0 (\mathfrak{a}^n m = 0)\}.$$

Assume that A is noetherian, and M is any A -module. Show that $\widetilde{\Gamma_{\mathfrak{a}}(M)} \cong \mathcal{H}_Z^0(\mathcal{F})$, where $Z = V(\mathfrak{a})$ and $\mathcal{F} = \widetilde{M}$.

- (e) Let X be a noetherian scheme, and let Z be a closed subset. If $\mathcal{F}$ is a quasi-coherent (resp., coherent) $\mathcal{O}_X$ -module, then $\mathcal{H}_Z^0(\mathcal{F})$ is also quasi-coherent (resp., coherent).

5.8. Again let X be a noetherian scheme, and $\mathcal{F}$ a coherent sheaf on X . We will consider the function

$$\varphi(x) = \dim_{k(x)} \mathcal{F}_x \otimes_{\mathcal{O}_X} k(x),$$

where $k(x) = \mathcal{O}_x/m_x$ is the residue field at the point $x \in X$. Use Nakayama's lemma to prove the following results.

- (a) The function φ is *upper semi-continuous*, i.e., for any $n \in \mathbb{Z}$, the set $\{x \in X \mid \varphi(x) \geq n\}$ is closed.
- (b) If $\mathcal{F}$ is locally free and X is connected, then φ is a constant function.
- (c) Conversely, if X is reduced and φ is constant, then $\mathcal{F}$ is locally free.

5.9. Let S be a graded ring, generated by S_1 as an S_0 -algebra, M a graded S -module, and $X = \text{Proj } S$.

- (a) Show there is a natural homomorphism $\alpha : M \rightarrow \Gamma_*(\widetilde{M})$.
- (b) Assume now that $S_0 = A$ is a finitely generated k -algebra for some field k , that S_1 is a finitely generated A -module, and that M is a finitely generated S -module. Show that the map α is an isomorphism in all large enough degrees, i.e., there is a $d_0 \in \mathbb{Z}$ such that for all $d \geq d_0$, $\alpha_d : M_d \rightarrow \Gamma(X, \widetilde{M}(d))$ is an isomorphism. [Hint: Use the methods of the proof of (5.19).]
- (c) With the same hypothesis, we define an equivalence relation $\approx$ on graded S -modules by saying $M \approx M'$ if there is an integer d such that $M_{\geq d} \cong M'_{\geq d}$. Here, $M_{\geq d} = \bigoplus_{n \geq d} M_n$. We say that a graded S -module is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\sim$ and Γ_* induce an equivalence of categories between the category of quasi-finitely generated graded S -modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$ -modules.

Proof. (a) Fix generators $\{x_i\}_{i \in I} \subset S_1$ of S as an S_0 -algebra. This gives us an affine open cover $\{D_+(x_i) \cong \text{Spec } S_{(x_i)}\}_{i \in I}$ of $\text{Proj } S$ such that

$$\begin{aligned} \widetilde{M}|_{D_+(x_i)}(n) &\cong \widetilde{M(n)}|_{D_+(x_i)} \\ &\cong (M(n)_{(x_i)})^\sim \\ &\cong ((M_{x_i})_n)^\sim \end{aligned}$$

since restriction commutes with tensor products, and $x_i \in S_1$ implies

$$(M \otimes_S N)_{(x_i)} = M_{(x_i)} \otimes_{S_{(x_i)}} N_{(x_i)}.$$

Then

$$\begin{aligned}
\Gamma_*(\widetilde{M}|_{D_+(x_i)}) &\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i)}(n)) \\
&\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(\operatorname{Spec} S_{(x_i)}, ((M_{x_i})_n)^\sim) \\
&\cong \bigoplus_{n \in \mathbb{Z}} (M_{x_i})_n \\
&\cong M_{x_i}
\end{aligned}$$

supplies us with a natural map $\alpha_i : M \hookrightarrow M_{x_i} \xrightarrow{\cong} \Gamma_*(\widetilde{M}|_{D_+(x_i)})$ for each i . Using a similar argument to the above, we define maps $\alpha_{ij} : M \hookrightarrow M_{x_i x_j} \xrightarrow{\cong} \Gamma_+(\widetilde{M}|_{D_+(x_i x_j)})$ and show that $\rho_{D_+(x_i x_j)} \circ \alpha_i = \alpha_{ij} = \rho_{D_+(x_i x_j)} \circ \alpha_j$.

Fixing $m \in M$, the $\alpha_i(m)_n \in \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i)}(n))$ glue together to a section $\alpha(m)_n \in \Gamma(X, \widetilde{M})$. Only finitely many $\alpha_i(m)_n$ are nonzero for each i , so only finitely many $\alpha(m)_n$ are nonzero, whence $\alpha(m) = \sum_n \alpha(m)_n \in \Gamma_*(\widetilde{M})$ is well-defined. We take $\alpha : m \mapsto \alpha(m)$ to be our natural map.

(b) By (I, 7.4), we have a finite filtration

$$0 = M^0 \subset M^1 \subset \cdots \subset M^r = M$$

of M by graded submodules, where for each i , $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogeneous prime ideals $\mathfrak{p}_i \trianglelefteq S$ and some integer n_i . Combining this with exactness of $\sim$, left exactness of $\Gamma(X, \cdot)$, and exactness of taking degree gives us the following diagram of S_0 -modules with exact rows, for any $d \in \mathbb{Z}$.

$$\begin{array}{ccccccc}
0 & \longrightarrow & M_d^{i-1} & \longrightarrow & M_d^i & \longrightarrow & M^i/M_d^{i-1} \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & \Gamma(X, \widetilde{M}^{i-1}(d)) & \longrightarrow & \Gamma(X, \widetilde{M}^i(d)) & \longrightarrow & \Gamma(X, \widetilde{M^i/M^{i-1}}(d))
\end{array}$$

If we have that the left and right vertical maps are isomorphisms, the five lemma implies the middle map is an isomorphism.

The result clearly holds for $M^0 = 0$, so it suffices to show the result for the $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n)$. Our map at degree d is an isomorphism iff it is a bijection, allowing us to disregard our original S and reduce to the case of $M = S$ a graded integral domain, finitely generated by S_1 as an S_0 -algebra, and $S_0 = A$ a finitely generated integral domain over k .

Take such S , with $x_0, \dots, x_r \in S_1$ generators as an A -module. Define $T_{\geq n} := \bigoplus_{d \geq n} T_d$ for any graded ring T , and denote $S' := \Gamma_*(\mathcal{O}_X)_{\geq 0}$. We will show that

$$\alpha_{\geq m} : S_{\geq m} \rightarrow S'_{\geq m}$$

is an isomorphism for $m \gg 0$. S is an integral domain, so the localization maps $S \rightarrow S_f$ are injective for all $f \neq 0$. In particular, each of the $S \rightarrow S_{x_i}$, $S_{x_i} \rightarrow S_{x_i x_j}$ are injective, as is $S \rightarrow S_{x_0 \cdots x_r}$. Thus, $\alpha_{d \geq 0} : S \rightarrow S'$ corresponds to the inclusion $S \subset S' \subset \bigcap_i S_{x_i}$. It suffices to show $S'_{\geq m} \subset S_{\geq m}$ for $m \gg 0$.

By the proof of 5.19, we have that S' is a finitely generated S -module. Take such a set of homogeneous generators $y_0, \dots, y_l$ of degree $d_j \geq 0$. Since $y_j \in S_{x_i}$ for all i , we can find a single integer n such that $x_i^n y_j \in S$ for all i . The x_i generate S_1 , so the monomials in the

x_i of degree m generate S_m for any m . Then we can take larger n such that $fy_j \in S$ for all $f \in S_n$. Since y_j is of positive degree, $fy_j \in S_{\geq n} := \bigoplus_{d \geq n} S_d$ whenever $f \in S_{\geq n}$. Take one n that works for all y_j , so that $S_{\geq n} \subset S_{\geq n}[y_0, \dots, y_l] \subset S_{\geq n}$. Then take $m = n + \max_j d_j$, so that

$$S'_{\geq m} \subset S_{\geq n}[y_0, \dots, y_l] = S_{\geq n} \subset S_{\geq m}.$$

We conclude that $\alpha_{\geq m} : S_{\geq m} \rightarrow S'_{\geq m}$ is an isomorphism for $m \gg 0$, as desired.

(c) If $M_d = 0$ for $d \gg 0$, then any $m \in \Gamma(X, \widetilde{M})$ restricts to $m_i \in M_{x_i}$ with $m = \frac{x_i^d m}{x_i^d} = 0$, whence $\widetilde{M} = 0$. Note that $\ker(\phi_d) = (\ker \phi)_d$ and $\text{cok}(\phi_d) = (\text{cok} \phi)_d$, so the following sequence is unambiguous and exact.

$$\ker \phi_{\geq d} \longrightarrow M_{\geq d} \xrightarrow{\phi_{\geq d}} M'_{\geq d} \longrightarrow \text{cok} \phi_{\geq d}$$

Then if $\phi_{\geq d} : M_{\geq d} \rightarrow M'_{\geq d}$ is an isomorphism for $d \gg 0$, exactness of $\sim$ gives us

$$0 = \widetilde{\ker \phi_{\geq d}} \longrightarrow \widetilde{M_{\geq d}} \xrightarrow{\phi_{\geq d}} \widetilde{M'_{\geq d}} \longrightarrow \widetilde{\text{cok} \phi_{\geq d}} = 0$$

is exact, whence $\widetilde{M} \cong \widetilde{M'}$. Thus, $\sim$ induces a map from $\approx$ -classes to quasi-coherent $\mathcal{O}_X$ -modules. Under the assumption that S is noetherian, $\sim$ induces a map from $\approx$ -classes of quasi-finitely generated S -modules to coherent $\mathcal{O}_X$ -modules.

We can use 5.19 to show that $\Gamma_*(\cdot)$ induces a map from coherent $\mathcal{O}_X$ -modules to $\approx$ -classes of quasi-finitely generated S -modules as well. By 5.15, we have a natural isomorphism $\beta : \Gamma_*(\cdot) \sim \rightarrow \text{id}$, and by (b) we have a natural isomorphism of $\approx$ -classes $\alpha : \text{id} \rightarrow \Gamma_*(\cdot)$. We conclude that $\Gamma_*(\cdot)$ and $\sim$ induce an equivalence of categories, as desired. $\square$

5.10. Let A be a ring, $S = A[x_0, \dots, x_r]$, and $X = \text{Proj } S$. We have seen that a homogeneous ideal I in S defines a closed subscheme of X (Ex. 3.12), and that convrsely every closed subscheme of X arises in this way (5.16).

(a) For any homogeneous ideal $I \subset S$, we define the saturation $\bar{I}$ of I to be

$$\bar{I} := \{s \in S \mid \forall 0 \leq i \leq r \ (\exists n > 0 \ (x_i^n s \in I))\}$$

We say that I is *saturated* if $I = \bar{I}$. Show that $\bar{I}$ is a homogeneous ideal of S .

- (b) Two homogeneous ideals I_1 and I_2 of S define the same closed subscheme of X iff they have the same saturation.
- (c) If Y is any closed subscheme of X , then the ideal $\Gamma_*(\mathcal{I}_Y)$ is saturated. Hence, it is the largest homogeneous ideal defining the subscheme Y .
- (d) There is a 1-1 correspondence between saturated ideals of S and closed subschemes of X .

Proof. (a) If $s \in \bar{I}$, then we can write $s = \sum_d s_d$ as the sum of its homogeneous parts, so that $x_i^n s = \sum_d x_i^n s_d \in I$ and I homogeneous implies $x_i^n s_d \in I$ for each d . We conclude that $s_d \in \bar{I}$ for all d , whence $\bar{I}$ is homogeneous.

(b) Fix homogeneous ideals $I_1, I_2 \leq S$. We have an affine cover $D_+(x_i) \cong \text{Spec } S_{(x_i)}$ of X , so it suffices to show that

$$\frac{S_{(x_i)}}{(I_1)_{(x_i)}} \cong \widetilde{S/I_1}|_{D_+(x_i)} \cong \widetilde{S/I_2}|_{D_+(x_i)} \cong \frac{S_{(x_i)}}{(I_2)_{(x_i)}}$$

for all $0 \leq i \leq r$ iff $\bar{I}_1 = \bar{I}_2$. This occurs iff the kernels of the respective projection maps are the same, i.e. $(I_1)_{(x_i)} = (I_2)_{(x_i)}$. Clearly this holds if $(I_1)_{x_i} = (I_2)_{x_i}$, so suppose conversely that the equality holds only in degree 0. Then x_i is a unit, so

$$\begin{aligned} f \in ((I_1)_{x_i})_d &\iff f/x_i^d \in (I_1)_{x_i} = (I_2)_{x_i} \\ &\iff f \in ((I_2)_{x_i})_d \end{aligned}$$

gives us that $(I_1)_{x_i} = (I_2)_{x_i}$.

We have that

$$S \cap (I_1)_{x_i} = \{s \in S : \exists n > 0 (x_i^n s \in I_1)\},$$

so

$$\bar{I}_1 = S \cap \bigcap_{i=0}^r (I_1)_{x_i} = S \cap \bigcap_{i=0}^r (I_2)_{x_i} = \bar{I}_2$$

whenever the equality of localizations holds. Conversely, suppose $(I_1)_{x_i} \neq (I_2)_{x_i}$ so, without loss of generality, there exists $s \in (I_1)_{x_i} \setminus (I_2)_{x_i}$. Then we can take $n > 0$ such that $x_i^n s \in I_1 \subset \bar{I}_1$, but $x_i^m(x_i^n s) \notin I_2$ for all $m > 0$ implies $x_i^n s \in \bar{I}_1 \setminus \bar{I}_2$, whence $\bar{I}_1 \neq \bar{I}_2$.

(c) We have that $\mathcal{I}_Y = \ker(\iota^\sharp : \mathcal{O}_Y \rightarrow i_*\mathcal{O}_X)$ is a subsheaf of $\mathcal{O}_Y$, so

$$J_i := \Gamma(D_+(x_i), \mathcal{I}_Y) \subset \Gamma(D_+(x_i), \mathcal{O}_Y) = S_{(x_i)}$$

implies

$$J_i(n) \cong \Gamma(D_+(x_i), \mathcal{I}_Y(n)) \subset \Gamma(D_+(x_i), \mathcal{O}_Y(n)) = S(n)_{(x_i)}$$

for all $n \in \mathbb{Z}$, whence $\Gamma_*(\mathcal{I}_Y) \subset \Gamma_*(\mathcal{O}_X) \cong S$ by 5.15.

Now, fix $s \in \bar{J}_m$, so that $x_i^r s \in J$ for $r \gg 0$. Write

$$x_i^m J_i = J_i(m) \subset S(m)_{(x_i)} = x_i^m S_{(x_i)}.$$

We have $s_i := s|_{D_+(x_i)} \in x_i^m S(m)_{(x_i)}$, and $x_i^r s_i \in (\Gamma_*(\mathcal{I}_Y)|_{D_+(x_i)})_{m+r} = J_{m+r}$. Then $x_i^r s_i = x_i^{m+r} j_i$ for $j_i \in J_i$, so $x_i \in J_i^\times$ gives us

$$s_i = x_i^r j_i \in J_m.$$

Then s is a gluing of sections in $s_i \in \Gamma(D_+(x_i), \mathcal{I}_Y(m))$, so $s \in \Gamma(X, \mathcal{I}_Y(m)) \subset \Gamma_*(\mathcal{I}_Y)$, as desired.

(d) To any saturated ideal $I \leq S$, we associated the closed subscheme $Y_I := \text{Proj } S/I$. By (b), this map $I \mapsto Y_I$ is injective. By (c), it is surjective, with $\Gamma_*(\mathcal{I}_Y) \mapsto \text{Proj } S/\Gamma_*(\mathcal{I}_Y) \cong Y$ for any closed subscheme Y . This gives us our bijection. $\square$

5.11. Let S and T be two graded rings with $S_0 = T_0 = A$. We define the *Cartesian product* $S \times_A T$ to be the graded ring $\bigoplus_{d \geq 0} S_d \otimes_A T_d$. If $X = \text{Proj } S$ and $Y = \text{Proj } T$, show that $\text{Proj}(S \times_A T) \cong X \times_A Y$, and show that the sheaf $\mathcal{O}(1)$ on $\text{Proj}(S \times_A T)$ is isomorphic to the sheaf $p_1^*(\mathcal{O}_X(1)) \otimes p_2^*(\mathcal{O}_Y(1))$ on $X \times_A Y$.

The Cartesian product of rings is related to the Segre embedding of projective spaces in the following way. If $x_0, \dots, x_r$ is a set of generators for S_1 over A , corresponding to a projective embedding $X \hookrightarrow \mathbb{P}_A^r$, and if $y_0, \dots, y_s$ is a set of generators for T_1 , corresponding to a projective embedding $Y \hookrightarrow \mathbb{P}_A^s$, then $x_i \otimes y_j$ is a set of generators for $(S \times_A T)_1$, and hence defines a projective embedding $\text{Proj}(S \times_A T) \hookrightarrow \mathbb{P}_A^N$, with $N = rs + r + s - 1$. This is just the image of $X \times Y \subset \mathbb{P}^r \times \mathbb{P}^s$ in its Segre embedding.

Proof. Take generators $\{x_i\}_{i \in I} \subset S_+$ and $\{y_j\}_{j \in J} \subset T_+$ of S and T as A -algebras. Then any element $s \in S_d$ can be written s $\square$

- 5.12. (a) Let X be a scheme over a scheme Y , and let $\mathcal{L}, \mathcal{M}$ be two very ample invertible sheaves on X . Show that $\mathcal{L} \otimes \mathcal{M}$ is also very ample. [*Hint:* Use a Segre embedding.]
- (b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms of schemes. Let $\mathcal{L}$ be a very ample invertible sheaf on X relative to Y , and let $\mathcal{M}$ be a very ample invertible sheaf on Y relative to Z . Show that $\mathcal{L} \otimes f^* \mathcal{M}$ is a very ample invertible sheaf on X relative to Z .

Proof. (a) If $\mathcal{L}, \mathcal{M}$ are very ample, then we have $i : X \rightarrow \mathbb{P}_Y^m$ and $j : Y \rightarrow \mathbb{P}_Z^n$ such that $\square$

- 5.13. Let S be a graded ring, generated by S_1 as an S_0 -algebra. For any integer $d > 0$, let $S^{(d)}$ be the graded ring $\bigoplus_{n \geq 0} S_n^{(d)}$, where $S_n^{(d)} = S_{nd}$. Let $X = \text{Proj } S$. Show that $\text{Proj } S^{(d)} \cong X$, and that the sheaf $\mathcal{O}(1)$ on $\text{Proj } S^{(d)}$ corresponds via this isomorphism to $\mathcal{O}_X(d)$.

This construction is related to the d -uple *embedding* in the following way. Denote $A = S_0$. If $x_0, \dots, x_r$ is a set of generators for S_1 , corresponding to an embedding $X \hookrightarrow \mathbb{P}_A^r$, then the set of monomials of degree d in the x_i is a set of generators for $S_1^{(d)} = D_d$. These define a projective embedding of $\text{Proj } S^{(d)}$ which is none other than the image of X under the d -uple embedding of $\mathbb{P}_A^r$.

- 5.14. Let A be a ring, and let X be a closed subscheme of $\mathbb{P}_A^r$. We define the homogeneous coordinate ring $S(X)$ of X for the given embedding to be $A[x_0, \dots, x_r]/I$, where I is the ideal $\Gamma_*(\mathcal{I}_X)$ constructed in the proof of (5.14). (Of course, if A is a field and X a variety, this coincides with the usual definition!) Recall that a scheme X is *normal* if its local rings are integrally closed domains. A closed subscheme $X \subset \mathbb{P}_A^r$ is *projectively normal* for the given embedding, if the homogeneous coordinate ring $S(X)$ is an integrally closed domain. Now assume that A is a finitely generated k -algebra for some field k , and that X is a connected, normal closed subscheme of $\mathbb{P}_A^r$. Show that for some $d > 0$, the d -uple embedding of X is projectively normal, as follows.

- (a) Let S be the homogeneous coordinate ring of X , and let $S' = \bigoplus_{n \geq 0} \Gamma(X, \mathcal{O}_X(n))$. Show that S is a domain, and that S' is its integral closure. [*Hint:* First show that X is integral. Then regard S' as the global sections of the sheaf of rings $\mathcal{S} = \bigoplus_{n \geq 0} \mathcal{O}_X(n)$ on X , and show that $\mathcal{S}$ is a sheaf of integrally closed domains.]
- (b) Use (Ex. 5.9) to show that $S_d = S'_d$ for all sufficiently large d .
- (c) Show that $S^{(d)}$ is integrally closed for sufficiently large d , and hence conclude that the d -uple embedding of X is projectively normal.
- (d) As a corollary of (a), show that a closed subscheme $X \subset \mathbb{P}_A^r$ is projectively normal if and only if it is normal, and for every $n \geq 0$ the natural map $\Gamma(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}(n)) \rightarrow \Gamma(X, \mathcal{O}_X(n))$ is surjective.

- 5.16. For $(X, \mathcal{O}_X)$ a ringed space and $\mathcal{F}$ a sheaf of $\mathcal{O}_X$ -modules, we define the *tensor algebra*, *symmetric algebra*, and *exterior algebra* of $\mathcal{F}$ by taking the sheaves associated to the associated presheaf.

- (a) Suppose that $\mathcal{F}$ is locally free of rank n . Then $T^r(\mathcal{F})$, $S^r(\mathcal{F})$ and $\wedge^r(\mathcal{F})$ are locally free, of ranks rn , $\binom{n+r-1}{n-1}$, and $\binom{n}{r}$ respectively.

- (b) Again, let $\mathcal{F}$ be locally free of rank n . Then the multiplication map
- 5.17. *Affine morphisms.* A morphism $f : X \rightarrow Y$ is *affine* if there is an affine open cover $\{V_i\}$ of Y such that $f^{-1}(V_i)$ is affine for each i .
- (a) Show that f is an affine morphism if and only if for each every open affine $V \subset Y$, $f^{-1}(V)$ is affine.
 - (b) An affine morphism is quasi compact and separated. Any finite morphism is affine.
 - (c) Let Y be a scheme, and let $\mathcal{A}$ be a quasi-coherent sheaf of $\mathcal{O}_Y$ -algebras. Show that there is a unique scheme X , and a morphism $f : X \rightarrow Y$, such that for every open affine $V \subset Y$, $f^{-1}(V) \cong \text{Spec } \mathcal{A}(V)$, and for every inclusion $U \hookrightarrow V$ of open affines of Y , the morphism $f^{-1}(U) \hookrightarrow f^{-1}(V)$ corresponds to the restriction homomorphism $\mathcal{A}(V) \rightarrow \mathcal{A}(U)$. The scheme X is called **$\text{Spec } \mathcal{A}$** .
 - (d) If $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_Y$ -algebra, then $f : X = \mathbf{Spec} \mathcal{A} \rightarrow Y$ is an affine morphism and $\mathcal{A} \cong f_* \mathcal{O}_X$. Conversely, if $f : X \rightarrow Y$ is an affine morphism, then $\mathcal{A} = f_* \mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_Y$ -modules, and $X \cong \mathbf{Spec} \mathcal{A}$.
 - (e) Let $f : X \rightarrow Y$ be an affine morphism, and let $\mathcal{A} = f_* \mathcal{O}_X$. Show that f_* induces an equivalence of categories from the category of quasi-coherent $\mathcal{O}_X$ -modules to the category of quasi-coherent $\mathcal{A}$ -modules (i.e., quasi-coherent $\mathcal{O}_Y$ -modules having the structure of an $\mathcal{A}$ -module.) [*Hint:* For any quasi-coherent $\mathcal{A}$ -module $\mathcal{M}$, construct a quasi-coherent $\mathcal{O}_X$ -module $\widetilde{\mathcal{M}}$, and show that the functors f_* and $\widetilde{}$ are inverse to each other.]

Proof. (a) It suffices to show that an affine morphism has this property. Take $f : X \rightarrow Y$ affine with open affine cover $\{V_i\}$ such that each $f^{-1}(V_i)$ is affine, and let $V \subset Y$ be open affine. Affine communication lemma gives us a cover of V by $D(g)$, $g \in \mathcal{O}_Y(V_i)$ which are distinguished open in both V and V_i for some i . Note that each $f^{-1}(D(g_\alpha)) = D(f^\#(V_i)(g))$ is open affine. Then we may assume $V = Y$ is affine, and that each V_i is a distinguished open set of Y .

We have a cover of X by open affines $U_i = f^{-1}(V_i)$

(b) Let $f : X \rightarrow Y$ be affine. Then, for any $V \subset Y$ affine open, $f^{-1}(V)$ is affine by (a), and thus quasi-compact. It follows from 3.2 that f is quasi compact.

Taking an affine cover V_i of Y , we have an affine cover $U_i := f^{-1}(V_i)$ of X . Then $f|_{U_i} : U_i \rightarrow V_i$ is a morphism of affine schemes, hence separated, so that $\Delta|_{U_i} : U_i \rightarrow U_i \times_{V_i} U_i$ is a closed immersion. It follows that $\Delta : X \rightarrow X \times_Y X$ is in fact a closed immersion, i.e. f is separated.

If f is finite, then there is an affine cover $V_i \cong \text{Spec } B_i$ of Y such that $f^{-1}(V_i) \cong \text{Spec } A_i$ such that A_i is a finitely generated B_i module. In particular, $f^{-1}(V_i)$ is affine.

(c) □

- 5.18. *Vector Bundles.* Let Y be a scheme. A (*geometric*) *vector bundle* of rank n over Y is a scheme X and a morphism $f : X \rightarrow Y$, together with additional data consisting of an open covering $\{U_i\}$ of Y , and isomorphisms $\psi_i : f^{-1}(U_i) \rightarrow \mathbb{A}_{U_i}^n$, such that for any i, j , and for any open affine subset $V = \text{Spec } A \subset U_i \cap U_j$, the automorphism $\psi = \psi_j \circ \psi_i^{-1}$ of $\mathbb{A}_V^n = \text{Spec } A[x_1, \dots, x_n]$ is given by a